

ASSIGNMENT

FoC

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CSE A

Roll No 25

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Q) A multinational company wants to connect offices across different countries. Identify the type of network & explain the key challenges.

ans) Wide area network (WAN)

A multinational company connecting offices across different countries would use a Wide Area network (WAN) — specifically likely a combination of MPLS (Multiprotocol Label switching), SD-WAN (Software-defined WAN), or VPN (Virtual Private Network), technologies over leased lines or the public internet.

Key challenges of a multinational WAN

1) Latency & distance

Data travelling b/w countries must cross thousands of km of undersea cables or satellite links. Round-trip times, which causes noticeable delays in real-time applications like VoIP, video conferencing & ERP sys.

2) Bandwidth & Cost

International leased lines (MPLS) are significantly more expensive than domestic circuits. Balancing adequate bandwidth against cost is a constant trade-off, & offices in regions with poor infrastructure (parts of Africa, South Asia) may have inherently limited connectivity options.

3) Security

Data traverses public internet segments, 3rd party carrier networks, & undersea ^{cables} ~~cities~~ — all potential interception points. End-to-end encryption (IPSec VPN or TLS), firewalls,

2

and zero-trust access policies become non-negotiable at every office.

- 4) Regulatory compliance
Different countries have different data sovereignty laws. The EU's GDPR, India's DPDP Act, & China's MLPS all impose restrictions on where data can be stored & how it can be transferred across borders. Routing traffic natively can inadvertently violate these laws.
- 5) Reliability & redundancy
A single WAN link failure can cut off an entire office. Multinational WANs need redundant paths — typically a primary MPLS circuit backed by a 2^o internet/SD-WAN path — along with failover automation to prevent extended outages.
- 6) Time zones & network management
Centralized monitoring & maintenance become complex when offices span 10+ time zones. Scheduled maintenance windows don't disrupt anyone are nearly impossible, & 24/7 NOC (Network Operations Centre) coverage becomes necessary.
- 7) Interoperability & standardisation
Each office may have acquired different hardware or ISP contracts over time. Harmonising routing protocols, QoS policies, & security standards across all sites requires careful governance & often a significant migration effort.

Q) Convert the IP address 192.168.10.14 to binary, Identify the network & host portions for a subnet mask of 255.255.255.0

192 $\rightarrow$ 11000000

168 $\rightarrow$ 10101000

10 $\rightarrow$ 1010

14 $\rightarrow$ 1110

Network portion (first 3 octets/24 bits): 192.168.10

Host portion (last octet/8 bits): 14

Q) Find the 1st address, last address & no of addresses for the IP address 192.208.243.37/16

• First address 192.208.0.0
(network)

• Last address 192.208.255.255
(broadcast)

• Total addresses 63,536

• Usable host range 192.208.0.1 - 192.208.255.254

• Usable hosts 63,534